

Water Interactions with Terminal Hydroxyls on TiO₂(110)[†]

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Received: April 23, 2010; Revised Manuscript Received: August 6, 2010

A combination of scanning tunneling microscopy and density functional theory has been used to investigate the interactions between water molecules and terminal hydroxyls (OH_t's) adsorbed on the TiO₂(110) surface at 300 K. We show that OH_t's have a significant effect on the water reactivity. Two distinctive reaction pathways are unraveled depending on whether H₂O and OH_t are on the same or adjacent Ti rows. The underlying reaction mechanisms involve proton transfer from H₂O to OH_t leading to the formation of new H₂O molecules, accompanied by O scrambling and along- or across-row apparent motion of OH_t's.

1. Introduction

Water over the rutile TiO₂(110) surface is one of the most studied model system for the fundamental studies of metal oxide surface reactivity and has attracted much attention.^{1–6} Furthermore, it plays an important role in various photocatalytic applications of titanium dioxide, including oxidation of organic pollutants and hydrogen production via water splitting.^{7–9} Studies of hydroxylation of the TiO₂(110) surface, reduced by annealing in ultrahigh vacuum (UHV), show that water preferentially dissociates at surface oxygen vacancy (V_O) sites, generating bridging hydroxyl groups (OH_b's).^{10–12} Recent experimental evidence also reveals that H₂O molecules at regular Ti sites are in a “pseudo-dissociated” state, rapidly switching between dissociated and molecular geometries that are in dynamical equilibrium.^{13–15} Water surface chemistry on TiO₂(110) can be significantly affected by the coadsorbates. For example, H₂O readily dissociates near oxygen adatoms (O_a's),^{13,16} that form upon O₂ oxidation of reduced TiO₂(110).^{16–21} In addition, while water is very mobile on the TiO₂(110) around 300 K,²² it can also induce motion of other coadsorbed species, e.g., the O_a's and OH_b's.^{13,22} However, in general the interactions of water with coadsorbates have received much less attention so far.

One of the most important coadsorbates is certainly oxygen, which is a common environmental reactant and also plays an important role in many photocatalytic processes.^{2,7,8} Reactions between O₂ and H₂O on catalytically active surfaces often involve intricate mechanisms with a number of possible surface-bound reactive intermediates.^{16,23–26} In our previous scanning tunneling microscopy (STM) study, we directly imaged for the first time Ti-bonded terminal hydroxyl (OH_t) species formed upon oxygen interaction with a partially hydroxylated TiO₂(110) surface at 300 K.²⁷ In a separate study, we have demonstrated that the OH_t's can also form as a result of H₂O interaction with

O_a's.¹³ We have also reported previously that in the case of O₂ interaction with a fully hydroxylated TiO₂(110), single OH_t's species have also been tentatively detected at the surface after large oxygen expositions.²⁶ In addition, (OH_t–H₂O) pair diffusion was recently imaged with STM at low temperatures (170–200 K).²⁸ Yet, the role of OH_t's in the reactivity of TiO₂(110) is largely unknown.

In this study, we investigate the interactions of water and terminal hydroxyls on TiO₂(110) surfaces at 300 K using high-resolution STM and density functional theory (DFT) calculations. We find that H₂O readily reacts with OH_t's following one of two distinctive reaction pathways that result in an apparent motion of OH_t's. Our DFT calculations support the experimental results and reveal underlying mechanisms of the observed processes.

2. Experimental Section

Experiments in this study were performed in two UHV-STM systems that have similar setups,^{18,29} thus only one is described in a detail here. This system (base pressure 3×10^{-11} Torr) is equipped with a variable-temperature STM (Omicron), a semi-spherical electron energy analyzer (Omicron), a mass spectrometer (Ametek), and electron and ion guns (VG and SPECS, respectively). The single-crystal rutile TiO₂(110)-(1 × 1) surface (Princeton Scientific) was prepared by multiple (~ few tens) cycles of Ar ion sputtering (2 keV) and UHV annealing (800–900 K), resulted in the initial V_O coverage of ~0.09 ML. At the beginning of each experiment, which was carried out at 300K, the sample was flash-annealed to 600 K. In order to form the OH_t's species, both O_a's and OH_b's should be present on the surface. Hence, we exposed a reduced TiO₂(110) surface to O₂ dosing through a dedicated doser, and H₂O either through another doser or by adsorption from UHV background (in both cases, the obtained results were similar). Coverages of surface species were determined by direct counting on the STM images and expressed in monolayer units (1 ML = 5.2×10^{14} atoms/cm²). We have analyzed the same surface area before and after adsorption, and this allowed us to monitor changes caused by the adsorption of individual molecules. STM tips were home-made from electrochemically etched W wire and cleaned in situ by the annealing and ion sputtering.³⁰ Presented STM (empty state) images were collected in a constant-current (~0.1 nA)

[†] Part of the “D. Wayne Goodman Festschrift”.

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mode at positive sample bias voltage of 1.5–1.8 V. The resulting images were processed using WSxM software.³¹

3. Computational Details

We performed DFT calculations using a slab model with three-dimensional periodic boundary conditions to model several reactions of interest over the TiO_2 (110) surface. A (4×2) supercell (13.12 Å by 11.96 Å), being four trilayers (or 12 atomic layers) thick, was used to describe the surface. We tested for vacuum space convergence by using vacuum spacings of 13 Å and 17 Å between slabs. The two vacuum spacings gave energies that agreed within 0.4 meV, indicating that the vacuum distances were sufficiently converged. Reciprocal space was treated by a single k-point, Γ , as our code currently only supports this k-point. We therefore ensured that the reciprocal space sampling density was converged by running calculations using larger supercells (or equivalently finer reciprocal space sampling density). We found energies for H-transfer processes using a (6×3) slab to agree within 2 meV of results using a (4×2) slab. All calculations were performed using the Perdew–Burke–Ernzerhof (PBE) exchange correlation functional.³² Generalized gradient approximation (GGA) functionals have been shown to accurately model H_2O – TiO_2 surface chemistry in many instances,^{13,19,33} despite their inability to correctly describe the band gap. Hybrid functional calculations are rather time-consuming and there is still no agreement as to the best approach for modeling TiO_2 with these functional, so we did not pursue using hybrid functionals. Core electrons were modeled by Goedecker–Teter–Hutter type pseudopotentials.^{34,35} Valence electrons were modeled by a dual basis set: plane waves to represent the electron density and Gaussian functions to represent the wave functions. This Gaussian and plane wave (GPW) method is implemented in the CP2K code.^{36–38} We utilized double- ζ Gaussian orbitals and plane waves up to a cutoff of 300 Ry.

Optimized geometries were converged to below 0.05 eV/Å. Activation barriers were calculated using the nudged elastic band (NEB) method with typically 7–8 images for each elementary process.³⁹ We did not perform vibrational analysis to fully confirm the transition states, since the minimal energy pathways were not very sharp, making exact transition state identification difficult. In such cases, identifying the exact transition state is computationally difficult. The NEB calculations however, as discussed below, indicate that the processes proceed readily at room temperature due to low barriers. A few select transition barriers were calculated using constrained optimization.

We assessed the accuracy of our computational methodology by comparing with results available in the literature. We calculated the lone H_2O adsorption energy over a stoichiometric surface to be -0.90 eV, similar to the -0.76 eV value calculated by Harris and Quong⁴⁰ and the -0.92 eV value calculated by Perron et al.³³ Our calculated OH_2O –Ti bond length was calculated to be 2.23 Å, very similar to the calculated distance of 2.25 Å reported in the literature.³³ These comparisons illustrate that our method generally agrees within 0.1 eV with available data (see also the computational results for H_2O adsorption over a reduced surface in the Supporting Information).

4. Results and Discussion

Figure 1 shows a typical STM image of the starting surface prepared by sequential exposure to H_2O and O_2 at 300 K to create OH_t species as discussed below and published previously.^{22,29} The TiO_2 (110) surface is composed of alternative rows

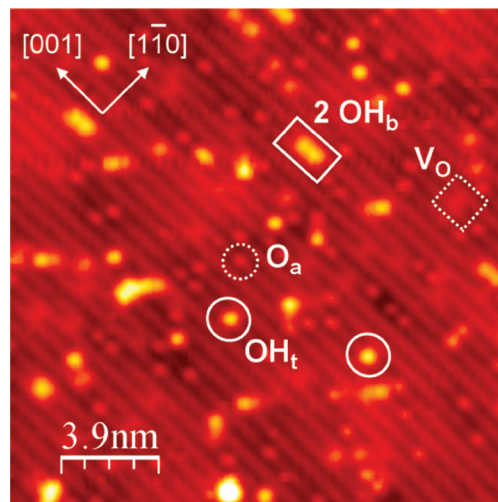


Figure 1. Empty-state STM image of a partially hydroxylated TiO_2 (110) surface after exposure to O_2 at 300 K. The initial V_O coverage on bare TiO_2 (110) was 0.09 ML.

of bridging oxygen (O_b) atoms and terminal Ti atoms oriented along the [001] direction. The STM image is dominated by electronic contrast, making the O_b rows appear dark and Ti rows bright.² Beyond the periodic surface structure, the unexposed TiO_2 (110) also contains V_O defects (Figure 1, marked by dotted square). Following the H_2O and O_2 exposure, additional small separated surface species are also present and marked in Figure 1. The brightest features on the O_b rows are OH_b 's (solid rectangular), which result from H_2O dissociation at V_O 's via following reaction, $\text{H}_2\text{O} + \text{V}_\text{O} + \text{O}_\text{b} \rightarrow 2 \text{OH}_\text{b}$.^{22,29} The small, bright features residing on the Ti rows (dotted circles) are O_a 's formed through O_2 dissociation either at V_O 's ($\text{O}_2 + \text{V}_\text{O} \rightarrow \text{O}_\text{b} + \text{O}_\text{a}$),^{17–19} or at regular Ti sites ($\text{O}_2 \rightarrow \text{O}_\text{a} + \text{O}_\text{a}$).^{20,21} Finally, the largest and brightest features located on Ti rows (marked by solid circles) are single OH_t species, which have been observed and identified in our recent STM work.²⁷ We have shown that a single OH_t can be formed as a result of hydrogen diffusion⁴¹ and proton transfer from OH_b to O_a ($\text{OH}_\text{b} + \text{O}_\text{a} \rightarrow \text{O}_\text{b} + \text{OH}_\text{t}$).²⁷ The pairs of OH_t 's can also be generated through H_2O interaction with O_a ($\text{H}_2\text{O} + \text{O}_\text{a} \rightarrow 2 \text{OH}_\text{t}$), but these paired OH_t 's are short-lived, because of the facile recombination process at 300 K leading to water reformation.¹³ In contrast, single OH_t species are stable and have a very low mobility at 300 K.²⁷ However, upon subsequent exposure of this surface to H_2O , we have observed a dramatic increase in the mobility of OH_t 's, as further discussed below.

4.1. Motion of OH_t Species across Bridging O Rows. To investigate the surface reactions between OH_t 's and H_2O , we have followed a particular area containing OH_t species before and after H_2O dose, as shown in Figure 2. From the comparison of marked regions one can see that the OH_t species has shifted to the adjacent Ti row on the right. The observation can be explained by considering the proton transfer from H_2O across the O_b row to OH_t . This mechanism implies that the process is mediated by water pseudodissociation ($\text{H}_2\text{O} \leftrightarrow \text{OH}_\text{b} + \text{OH}_\text{t}$).^{13,14,22} It has been shown recently that at 300 K H_2O on TiO_2 (110) is indeed in a dynamic equilibrium between the molecular and dissociated states.^{13,14,22} The separate reaction steps are illustrated schematically in the ball models in Figure 2, whereas the model in the dotted box schematically shows an assumed initial event of H_2O pseudodissociation and formation of short-lived intermediate state ($\text{OH}_\text{t} + \text{OH}_\text{b} + \text{OH}_\text{t}$). If the newly formed OH_t species could diffuse away, the H_2O reformation reaction would

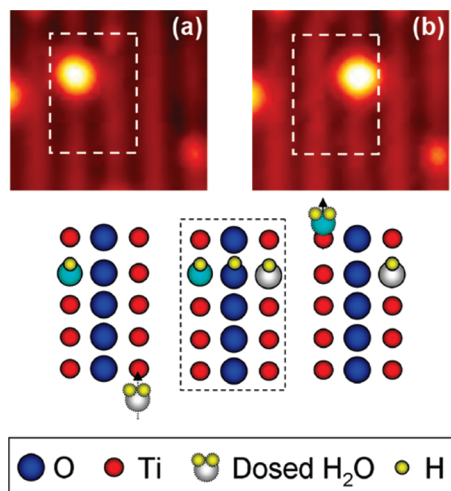


Figure 2. Two consecutive STM images (sampling rate 0.5 frame/min) of the same (3×3) nm² area after H₂O exposure showing the apparent OH_i motion across an O_b row, as a result of the interaction with H₂O at 300 K. Corresponding ball-models of the rectangular region marked in (a) and (b) illustrate the reaction path.

not occur. However OH_b and OH_i diffusion at 300 K is very slow and does not provide for an efficient OH_i separation.²⁷ Hence, the second proton transfer from the newly created OH_b to OH_i will lead to H₂O reformation (OH_b + OH_i → O_b + H₂O), as observed in our prior study.²⁷ This proton transfer from OH_b can result either in the formation of a new water molecule, leaving behind an OH_i that originated from the initial H₂O (schematically shown by ball models in Figure 2) or in the reformation of the original H₂O (not shown). The former case results in oxygen scrambling and OH_i across-row “motion”, while the latter case leaves the OH_i in its original position. Finally, the reaction is completed by H₂O diffusion, which at 300 K is too fast to be followed with STM.²² As a side comment, note that the oxygen scrambling also facilitates an apparent transfer of an H₂O molecule from one Ti row to another. Also note, that in many aspects the interaction between OH_i and H₂O on adjacent rows is similar to the previously observed H₂O interaction with O_a that resides on an adjacent row.¹³

Particular care has been taken to evaluate possible tip-induced effects on OH_i mobility. We have compared scanning at different bias voltages and tunneling currents, and also compared two areas of the same surface, whereas one has been scanned once and the other about 10–20 times more. No changes in the OH_i

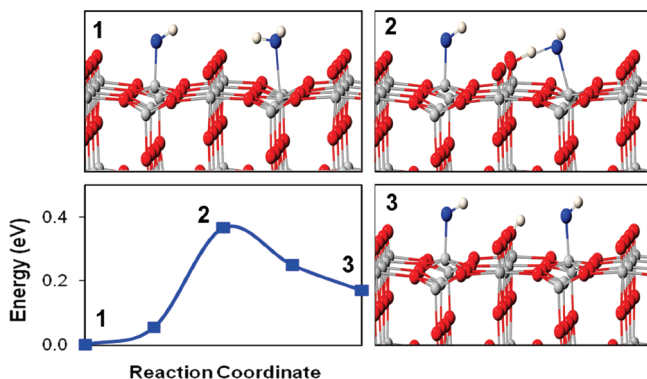


Figure 3. Reaction energy profile and selected intermediate states for H transfer between a H₂O molecule and OH_i on the neighboring Ti row (OH_i + H₂O ↔ OH_i + OH_b + OH_i). Ti atoms are shown in gray, O in red and blue, and H in light gray.

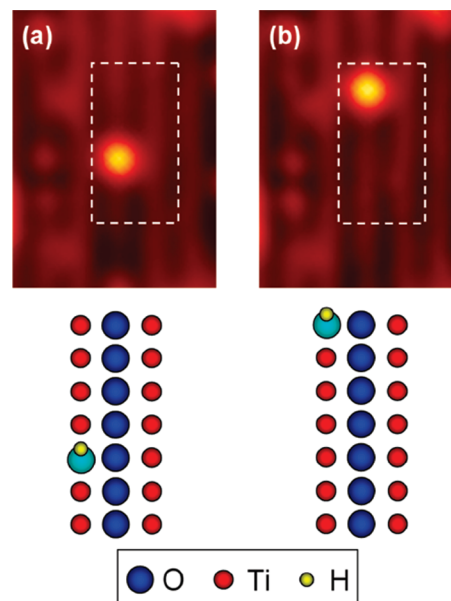


Figure 4. Two consecutive STM images of the same (3.4×4.9) nm² area showing OH_i motion (by four lattice constants) along the Ti row at 300 K, as a result of its reaction with H₂O.

dynamics have been observed in either case, indicating that tip effects were negligible under our experimental conditions.

The interpretation of the STM data presented in Figure 2 is supported by DFT calculations. Figure 3 shows the energy profile and selected intermediate states for H₂O dissociation in the presence of OH_i species on the adjacent Ti row. The calculations indicate that across-row proton transfers from H₂O to O_b and then to OH_i occur readily at 300 K, in agreement with the STM results. Specifically, the activation barrier for the initial proton transfer from H₂O to O_b is 0.37 eV in the presence of OH_i, similar to 0.39 eV, calculated for H₂O dissociation on a stoichiometric surface (not shown). The next proton transfer process from OH_b to OH_i to reform H₂O can proceed in either direction due to the symmetry of this configuration with an activation barrier of 0.20 eV. Since the calculated barriers for both proton transfers are low, these processes are very facile at 300 K. Thus, the H exchange between these two states, OH_i + H₂O ↔ OH_i + OH_b + OH_i, can occur many times, until water diffuses to the next Ti site and terminates the reactions. Note, that the (H₂O + OH_i) configuration is 0.17 eV more stable than (OH_i + OH_b + OH_i) configuration. Assuming Arrhenius behavior, under equilibrium conditions such an energy difference would on average result in the relative (OH_i + OH_b + OH_i) population being only ~0.2%. Also note that observation of the across-row OH_i mobility may be viewed as further evidence of the spontaneous water dissociation on Ti sites.

4.2. Motion of OH_i Species along Ti Rows. We have also observed high OH_i mobility along the same Ti row in the presence of H₂O. Remarkably, rather long OH_i hops of up to seven lattice constants have been detected in this case. An example of the along-row OH_i diffusion is illustrated in two consecutive STM images shown in Figure 4. From a comparison of Figure 4, panels a and b, the OH_i is found to hop four lattice constants (in this particular case) from its original position. With the limited available data, no conclusive travel distance distribution could be attained, while practically no one-unit-cell hops have been observed.

To identify the mechanism of the long-range motion of OH_i species along Ti rows, we have resorted to DFT calculations. Figure 5 shows the energy profile and selected intermediates

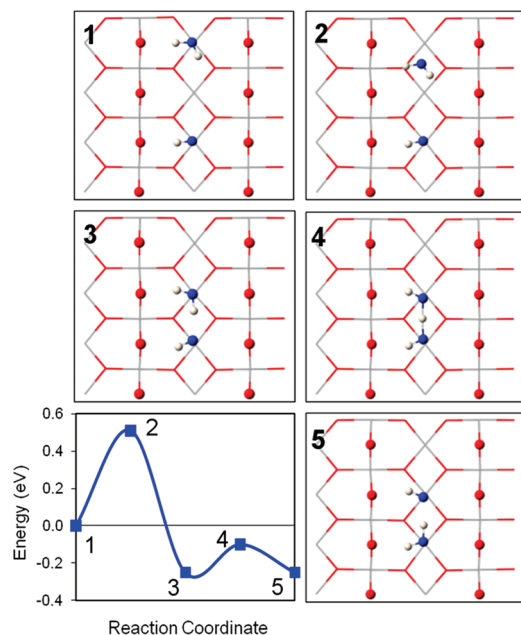


Figure 5. Reaction energy profile and selected intermediate states for water diffusion along a Ti row, H₂O dissociation, and subsequent proton transfer to OH_t.

along the reaction path involving H₂O diffusion toward OH_t (snapshots 1–3), followed by H₂O dissociation and proton transfer to OH_t (3–5). The calculated energy barriers are low for both diffusion and dissociation (0.51 and 0.15 eV, respectively), indicating that both processes are facile at 300 K. The proton transfer also results in the formation of a new H₂O molecule and an apparent OH_t shift one lattice constant up. However, such process can not facilitate further OH_t motion along the same direction. In contrast, since reverse proton transfer (5 → 3) also has a small barrier (0.15 eV) fast exchange between these two states, H₂O + OH_t ↔ OH_t + H₂O, is expected. The calculations suggest that H₂O becomes effectively trapped near the OH_t, with the activation energy barrier for water to diffuse away from the OH_t being ~0.76 eV. The later also correlates with the adsorption energy gain of 0.25 eV upon bringing H₂O and OH_t together, which is close to a typical H bond strength. Of course, the net migration of OH_t species across several sites may be accomplished by consecutive interactions with several H₂O molecules, following the mechanism described above, or by interaction with water dimers.⁴² However, this would require rather a high concentration of H₂O, and so it is unlikely to occur in the current studies. Otherwise, high H₂O concentration would also result in multiple across-row OH_t hops that have not been observed.

On the basis of the experimental evidence and theoretical calculations described above, we speculate that the observed long-range OH_t motion is a result of fast diffusion of H₂O + OH_t pairs. As already discussed (see Figure 5), the H₂O + OH_t ↔ OH_t + H₂O reaction allows only for a single site motion of OH_t's along one direction while multiple site hops are observed in Figure 4. To accomplish multiple site hops, H₂O molecule has to diffuse over the OH_t, effectively reforming the starting configuration shown in snapshot 3 of Figure 5.

Using DFT we have determined the following low-energy reaction path for the water hopping over the OH_t, as illustrated in Figure 6. In the starting configuration (snapshot 1), H₂O and OH_t reside on neighboring Ti sites, with a hydrogen bond between them formed through one of the H₂O protons. Next, both species rotate to the configuration, where a new H-bond

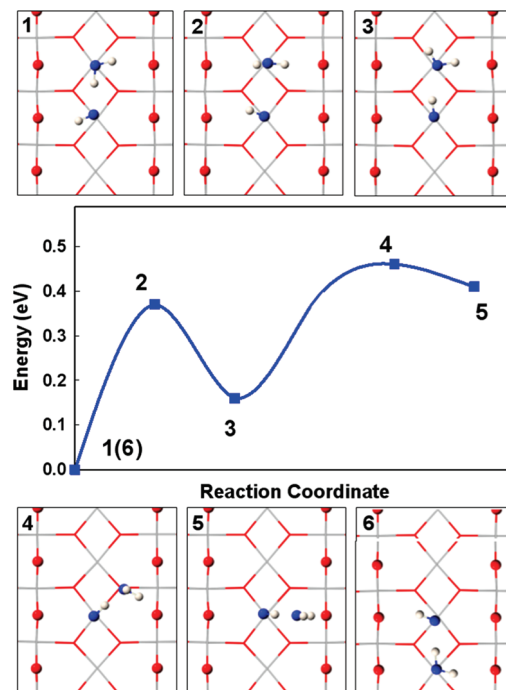


Figure 6. Reaction energy profile and selected intermediate states for H₂O hopping over OH_t.

is formed, with the proton provided by OH_t (snapshot 3). The process is slightly energetically uphill (0.16 eV) and has a barrier of 0.37 eV, through the transition state shown in the snapshot 2. The process of concerted rotation is important because it sets out further H₂O hopping. Subsequently, H₂O hops over the adjacent in-plane O atom (snapshot 4), and then to a new configuration (snapshot 5), where an additional H-bond between H₂O and nearest O_b likely stabilizes this intermediate state. Importantly, during this motion, the H-bond between the OH_t and H₂O is maintained and assists H₂O hop. The transition state region is rather flat which makes it hard to unequivocally define the exact configuration of transition state. The calculated activation barrier for this state is ~0.3 eV. After passing this symmetric midpoint (snapshot 5), the H₂O can continue moving in a same way until it reaches the final configuration (snapshot 6) with the H₂O on the “other” side of the OH_t. Such H₂O rollover motion, where intramolecular H bond formation plays an important role, resembles the mechanism recently reported for water dimer diffusion along the Ti row.⁴² The authors have shown there that H₂O dimer diffusion is enhanced when such rollover motion occurs, while here we demonstrate that OH_t diffusion is also enhanced by similar rollover motion.

The resulting activation energy barrier for H₂O diffusion over the OH_t, obtained from the energy profile in Figure 6, is 0.46 eV, demonstrating that this process occurs readily at 300 K. Note, that direct H₂O hopping over the OH_t is unfavorable, with the activation energy barrier of 0.89 eV (not shown). Also note, that besides the H₂O hopping over the OH_t as described above, the water diffusion from one Ti row to another over a similar, H-containing OH_b species has also been observed.²²

Repeated steps of proton transfer and H₂O hopping over OH_t, described in Figures 5 and 6, likely constitute the diffusion pathway for the observed long-range, water-mediated OH_t mobility. Generally, this mechanism can also be considered as diffusion of the (OH_t + H₂O) pairs. Such (OH_t + H₂O) pair diffusion (there called OH_tOH₂) was recently imaged with STM at low temperatures (170–200 K)²⁸ and, therefore, is expected to occur more readily at 300 K.

We also considered the effect of the presence of O vacancies in our DFT calculations by removing one O_b per unit cell to create a surface vacancy ratio of 1/8. In our previous work we indicated that H transfer barriers between O_a and OH_i were not affected by the presence of excess electrons.⁴³ We calculated H transfer reaction energy barriers over reduced surface and found them to agree with ones over stoichiometric surface within 0.02 eV (H transfer between H_2O and O_b) and 0.12 eV (H transfer between H_2O and OH_i), as shown in Figures S1 and S2 of the Supporting Information, respectively. For H_2O hopping over OH_i reaction pathway with V_O nearby, the barrier drops only slightly (0.06 eV, Figure S3). (Note that this pathway is not very symmetrical due to the presence of the V_O and is relatively flat, making an exact identification of the transition state rather difficult). However, while excess electrons can affect surface chemistry, in the current case H transfer reactions involving water and related species were not greatly affected, and our conclusions from the DFT calculations are the same regardless of whether stoichiometric or reduced surfaces were simulated.

5. Summary

We have performed a combined experimental and theoretical investigation of the reaction of molecular water with terminal hydroxyls on $TiO_2(110)$ at 300 K, and extracted molecular-level details about the underlying reaction mechanisms. By tracking the same surface area with high-resolution STM before and after water exposure, we have demonstrated that there are two distinctive reaction pathways involving multiple proton transfers. For water interaction with OH_i on an adjacent Ti row, the proton can be transferred through bridging oxygen to OH_b , which leads to the formation of a new water molecule and apparent across-row motion of OH_i due to O scrambling. This process further manifests the existence of the equilibrium between molecular and dissociated states of water on $TiO_2(110)$. If H_2O interacts with OH_i along the same Ti row, fast multistep OH_i motion along the Ti row is observed. Our DFT results show that this process is caused by the fast diffusion of $(OH_i + H_2O)$ pairs, whereby the underlying mechanism involves proton transfer and H_2O hopping over OH_i .

Acknowledgment. We thank M. A. Henderson, G. A. Kimmel, and N. G. Petrik for stimulating discussions. This work was supported by the U.S. Department of Energy (DOE) Office of Basic Energy Sciences, Division of Chemical Sciences, and performed at EMSL, a national scientific user facility sponsored by the DOE's Office of Biological and Environmental Research and located at PNNL. Computational resources were provided by the National Energy Research Scientific Computing Center and the EMSL.

Supporting Information Available: Computational results for H_2O adsorption, and energy profiles/selected intermediate states for reactions on reduced $TiO_2(110)$ surface. This information is available free of charge via the Internet at <http://pubs.acs.org>.

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JP1036876